

# Varsini Sakthivadivel Ramasamy

vsakthi1@jh.edu | varsini-sakthi.github.io | linkedin.com/in/varsini-sakthi | github.com/Varsini-Sakthi | 443-802-2041

## PROFESSIONAL SUMMARY

Bioinformatics MS graduate (JHU) pursuing an AI/ML PhD, specializing in deep learning and representation learning for high-dimensional biological data. Designed and trained a novel transformer architecture (SpatialFormer) and a variational generative model (scVI) from scratch; author of a manuscript in preparation on transcriptional heterogeneity modeling. Combines rigorous experimental benchmarking with domain expertise in single-cell and spatial genomics.

## EDUCATION

**Johns Hopkins University** – Baltimore, MD

Aug 2024 – May 2026

Master of Science in Bioinformatics

GPA: 3.89/4.0

• **Coursework:** Genomic Data Science; Algorithms for Bioinformatics; Applied ML; Statistics; Gene Expression Analysis & Visualization

**VIT University** – Vellore, India

Sep 2020 – May 2024

Bachelor of Technology in Biotechnology

GPA: 3.96/4.0

## EXPERIENCE

**Research Assistant:** Neuroimmunology & Single-Cell Genomics

Jan 2026 – Present | Johns Hopkins

- Built an end-to-end pseudobulk RNA-seq pipeline analyzing 20 quality-filtered samples across five developmental stages (E17–P60), integrating PCA, z-scored gene module scoring, Welch's t-test differential expression with Benjamini-Hochberg FDR correction, and principal-curve pseudotime trajectory inference; first-author manuscript in preparation.
- Defined four a priori microglial transcriptional programmes (homeostatic, phagocytic, NRF2/Hmox1 oxidative stress-responsive, and Apoc1-associated) from curated marker gene sets, and used pseudotime modeling to show the NRF2/Hmox1 programme peaks at intermediate developmental stages, revealing a transitional regulatory role in microglial maturation relevant to neuroinflammation and perinatal brain injury.

**Research Assistant:** scRNA-seq Representation Learning for Cell-Type Classification

Dec 2025 – Apr 2026 | Johns Hopkins

- Designed a controlled benchmark isolating representation quality as the only variable: trained a 30-D scVI variational autoencoder against a 30-component PCA baseline on 12,012 PBMC cells (36,601 genes, 2,000 HVGs), holding classifier architecture, train/test split, and random seed identical across both arms.
- Annotated 12 immune cell populations via Leiden clustering and marker-gene differential expression (Wilcoxon test), then evaluated both representations with logistic regression and an MLP; PCA outperformed scVI consistently across every metric (LR: 95.1% vs. 91.4%; MLP: 89.5% vs. 88.0%; 5-fold CV: 88.6% vs. 86.1%), showing added model complexity did not translate to better downstream performance on this single-batch dataset.

**Research Assistant:** Applied ML for Medical Imaging

Jun 2025 – Mar 2026 | Johns Hopkins

- Built a multiclass brain tumor classification pipeline on 3,064 MRI slices from 233 patients (meningioma, glioma, pituitary tumor; Cheng 2017 benchmark dataset), extracting first-order and GLCM texture radiomic features from tumor-masked regions and training Logistic Regression, Random Forest, and XGBoost with RandomizedSearchCV (20 iterations, stratified 5-fold CV) optimized for macro-F1 to counter class imbalance.
- Achieved 97.4% accuracy with XGBoost (Random Forest: 95.5%, tuned via CV; Logistic Regression: 92.9%, macro-ROC-AUC 0.92); applied SHAP to show GLCM texture descriptors of tumor heterogeneity drove classification, aligning with established radiological malignancy markers.
- Reported macro-averaged precision/recall (0.59/0.63) alongside accuracy to surface reduced performance on minority tumor classes (one with a single test example) rather than letting a high headline accuracy mask real limitations on rarer, clinically urgent subtypes.

**Teaching Assistant:** Practical Computer Concepts

Aug 2025 – May 2026 | Johns Hopkins

- Supported instruction in UNIX, SQL, and Python for Practical Computer Concepts for Bioinformatics; developed automated RNA-seq and proteomics analysis pipelines in Python and R to aid course demonstrations and student workflows.

## PUBLICATIONS

- ▷ Sakthivadivel Ramasamy V., Ozen M. *Homeostatic, phagocytic, NRF2/Hmox1 oxidative, Apoc1 and chemokine microglia transcriptional programmes in naïve mouse cerebral cortex from embryo to adulthood*. Manuscript in preparation.
- ▷ Antony A., Sakthi V., Veerappapillai S., Karuppasamy R. *Genome mining of secondary metabolites from Streptomyces spp. for the screening of novel and potent fungicide: Focus on rice blast disease*. Research Journal of Chemistry and Environment 2025
- ▷ Varsini S.R., Bhor S.S. *An Integrated Study to Extrapolate the Interaction of NMDAR with Potential Ligands for the Treatment of Alzheimer's Disease Symptoms*. International Journal of Biochemistry and Biomolecule Research 2023

## PROJECTS

---

### SpatialFormer: A Spatially Aware Transformer for Cell State Prediction

Apr 2026 – May 2026 | Johns Hopkins

- Designed a transformer encoding each Visium spot jointly with its 6 nearest spatial neighbors (2-layer, 4-head encoder, 8-token sequence), summing expression, coordinate, and token-type embeddings so attention weights over neighbors are learned rather than hard-coded.
- Achieved 89.1% validation accuracy (0.88 macro-F1) on 10x mouse brain data (2,796 Visium spots, 2,000 HVGs); a logistic-regression baseline outperforms it at 92.3%, framing the model as an interpretability contribution rather than a leaderboard result.
- Showed via attention analysis that the model allocates disproportionate weight to spatial neighbors specifically at anatomical boundaries (up to 54% neighbor attention vs. 2% in homogeneous tissue), a distinction a purely expression-based model cannot produce, and validated gene-saliency maps against canonical cell-type markers (Mbp, Gfap, Pcp4); deployed for public inference on HuggingFace Spaces.

## SKILLS

---

- **Machine Learning:** Supervised/unsupervised learning, Random Forests, representation learning, variational autoencoders (VAE), XGBoost, Deep learning (PyTorch), hyperparameter tuning, LLMs, model evaluation (ROC-AUC), feature engineering (SHAP)
- **Programming:** Python (pandas, scikit-learn, Biopython), R (tidyverse, Bioconductor), SQL, Bash, Linux/Unix, MatLab, AWS, Jupyter, Git.
- **Bioinformatics & Genomics Tools:** RNA-seq analysis, differential gene expression, single-cell RNA-seq (Scanpy, scVI), cell type annotation, transcriptomic data analysis, gene regulatory network analysis, NGS data analysis (Trimmomatic, Bowtie2/STAR), AutoDock, PyRx, GROMACS, Seurat, Discovery Studio, GEO, UCSC Genome browser, AlphaFold, MODELLER, Galaxy, ExPASy, ProSA-web.
- **Data Science & Visualization:** Data pipelines, Statistical modeling, exploratory data analysis (EDA), multi-omics data analysis, high-throughput biological data analysis, pipeline development, gene regulatory network analysis, Matplotlib, Seaborn, ggplot2.